

Family	Model	Comparison	AUC (first vs second)	AUC difference (95% CI)	P	P (BH)
Between omics layers	EN omics	Microbiome vs Metabolomics	0.491 vs 0.622	-0.130 (-0.212 to -0.047)	0.002	0.002
Between omics layers	EN omics	Microbiome vs Proteomics	0.491 vs 0.660	-0.169 (-0.250 to -0.085)	<0.001	<0.001
Between omics layers	EN omics	Metabolomics vs Proteomics	0.622 vs 0.660	-0.038 (-0.100 to +0.017)	0.198	0.198
Between omics layers	XGB omics	Microbiome vs Metabolomics	0.483 vs 0.661	-0.178 (-0.257 to -0.096)	<0.001	<0.001
Between omics layers	XGB omics	Microbiome vs Proteomics	0.483 vs 0.649	-0.166 (-0.246 to -0.086)	<0.001	<0.001
Between omics layers	XGB omics	Metabolomics vs Proteomics	0.661 vs 0.649	+0.012 (-0.042 to +0.068)	0.664	0.664
Between omics layers	RF clinical + EN	Microbiome vs Metabolomics	0.616 vs 0.654	-0.038 (-0.070 to -0.008)	0.014	0.021
Between omics layers	RF clinical + EN	Microbiome vs Proteomics	0.616 vs 0.666	-0.050 (-0.088 to -0.016)	0.006	0.017
Between omics layers	RF clinical + EN	Metabolomics vs Proteomics	0.654 vs 0.666	-0.012 (-0.048 to +0.025)	0.523	0.523
Between omics layers	RF clinical + XGB	Microbiome vs Metabolomics	0.622 vs 0.659	-0.037 (-0.073 to +0.003)	0.061	0.182
Between omics layers	RF clinical + XGB	Microbiome vs Proteomics	0.622 vs 0.647	-0.025 (-0.085 to +0.032)	0.409	0.614
Between omics layers	RF clinical + XGB	Metabolomics vs Proteomics	0.659 vs 0.647	+0.012 (-0.042 to +0.063)	0.658	0.658
Versus the clinical model	EN omics	Microbiome vs Clinical	0.491 vs 0.634	-0.142 (-0.222 to -0.054)	0.001	0.007
Versus the clinical model	XGB omics	Microbiome vs Clinical	0.483 vs 0.634	-0.151 (-0.229 to -0.068)	<0.001	0.004
Versus the clinical model	RF clinical + EN	Microbiome vs Clinical	0.616 vs 0.634	-0.018 (-0.033 to -0.002)	0.023	0.090
Versus the clinical model	RF clinical + XGB	Microbiome vs Clinical	0.622 vs 0.634	-0.011 (-0.035 to +0.011)	0.351	0.601
Versus the clinical model	EN omics	Metabolomics vs Clinical	0.622 vs 0.634	-0.012 (-0.089 to +0.065)	0.762	0.762
Versus the clinical model	XGB omics	Metabolomics vs Clinical	0.661 vs 0.634	+0.027 (-0.037 to +0.099)	0.446	0.635
Versus the clinical model	RF clinical + EN	Metabolomics vs Clinical	0.654 vs 0.634	+0.021 (-0.009 to +0.050)	0.169	0.366
Versus the clinical model	RF clinical + XGB	Metabolomics vs Clinical	0.659 vs 0.634	+0.026 (-0.013 to +0.065)	0.183	0.366
Versus the clinical model	EN omics	Proteomics vs Clinical	0.660 vs 0.634	+0.026 (-0.045 to +0.098)	0.476	0.635
Versus the clinical model	XGB omics	Proteomics vs Clinical	0.649 vs 0.634	+0.015 (-0.059 to +0.084)	0.682	0.744
Versus the clinical model	RF clinical + EN	Proteomics vs Clinical	0.666 vs 0.634	+0.033 (-0.002 to +0.069)	0.068	0.205
Versus the clinical model	RF clinical + XGB	Proteomics vs Clinical	0.647 vs 0.634	+0.014 (-0.047 to +0.074)	0.652	0.744